

# Acoustic-based drone detection systems using machine learning to enhance airspace security in defense and airport environments.

Aerospace and defense industry × Drone detection and airspace protection × Machine learning · OS090 v1 · refreshed 2026-08-18

| ATTRACTIVENESS                   | RIGHT TO WIN                         | COMPETITION                              | SERVICEABLE MARKET                            | HORIZON                                         | PORTFOLIO               |
|----------------------------------|--------------------------------------|------------------------------------------|-----------------------------------------------|-------------------------------------------------|-------------------------|
| <b>83</b><br>is the world moving | <b>58</b><br>can we play, can we win | <b>MEDIUM</b><br>1 named in the evidence | <b>€12.1m</b><br>modelled evidence · per year | <b>NEXT</b><br>regulation adopted or propose... | <b>L0</b><br>14 signals |

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

**Evidence gap (SC-13).** Orange has 0.0 published reference(s) in Aerospace and defense industry, below the threshold of 5. A customer conversation here has no proof point behind it yet.

## The opportunity

Acoustic-based drone detection using machine learning offers a new way to secure airspace around airports and defense sites, complementing camera-based systems that fail in fog or behind obstacles. This opportunity is timely as drone threats are forcing a fresh look at airport security, and Orange can deliver a sovereign, integrated service built on its Orange Drone Guardian offer and AI/defense expertise.

## Market opportunity

| Method                                             | Total (TAM)                     | Serviceable (SAM)               | Obtainable (SOM)              | Confidence |
|----------------------------------------------------|---------------------------------|---------------------------------|-------------------------------|------------|
| Bottom-up: enterprises x adoption x contract value | <b>€14.6m</b><br>€3.7m – €57.8m | <b>€12.1m</b><br>€3.1m – €47.7m | <b>€422k</b><br>€108k – €1.7m | modelled   |
| Observed: tendered contract value, annualised      | €45.7m                          | €45.7m                          | €1.6m                         | observed   |

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

### Bottom-up: enterprises x adoption x contract value

| Factor                                                    | Value                                                           | Basis      | Source                                        | As of          |
|-----------------------------------------------------------|-----------------------------------------------------------------|------------|-----------------------------------------------|----------------|
| Enterprises in Aerospace and defense industry (EU27_2020) | 3,354 enterprises                                               | observed   | Eurostat · sbs_sc_oww                         | 2024           |
| Enterprises adopting Machine learning                     | 7.8 % of enterprises (10+ employees)                            | observed   | Eurostat · isoc_eb_ain2 · E_AI_TML            | 2025           |
| Annual value of one engagement                            | €240k per enterprise per year                                   | assumption | config/sizing.yaml · fallback_bands_eur       | size-2026-08-a |
| Size-weighted buyer base                                  | 780 enterprise-equivalents at large-enterprise engagement value | assumption | config/sizing.yaml · size_class_value_weights | size-2026-08-a |
| Assumed obtainable share of the serviceable market        | 3.5 % of SAM                                                    | assumption | config/sizing.yaml · obtainable_share         | size-2026-08-a |

**Read this before quoting the number.** Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

### Observed: tendered contract value, annualised

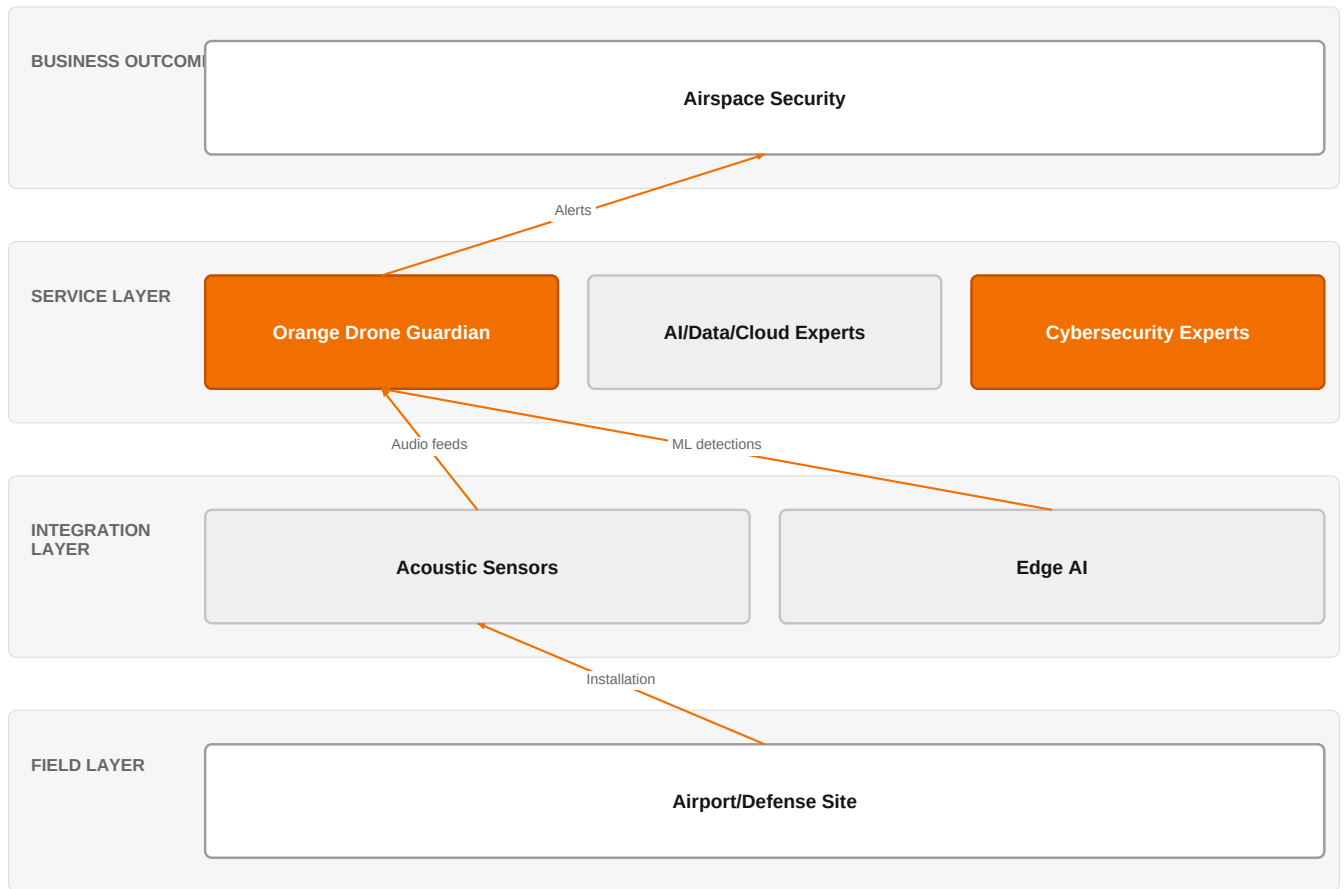
| Factor                                             | Value           | Basis      | Source                                | As of                  |
|----------------------------------------------------|-----------------|------------|---------------------------------------|------------------------|
| Tendered contract value in matching CPV categories | €45.7m per year | observed   | TED (Tenders Electronic Daily) · ted  | 2026-07-07..2026-07-08 |
| Assumed obtainable share of the serviceable market | 3.5 % of SAM    | assumption | config/sizing.yaml · obtainable_share | size-2026-08-a         |

**Read this before quoting the number.** Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 3 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in

this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

## The solution, in one picture

### Acoustic Drone Detection Service



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange integrates acoustic sensors and edge AI into a managed service that delivers airspace security alerts.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

## Who buys, and why now

The primary buyer is the head of airport operations or the chief security officer at an airport, who feels the pain of unauthorized drone incursions disrupting flights and threatening safety. The trigger is a specific incident—like the Ben Gurion incursion—or a regulatory mandate to install detection systems by a certain date, as seen in Vietnam. Defense agencies also buy for base protection, where the pain is the inability of cameras to detect drones in adverse conditions.

Sources: SIG-CBC989FFDC45 JFeed 2026-08-11; SIG-ABA568B306B8 Qazinform 2026-06-16

## Why it is hot now — every claim bound to a dated source

- A new drone detection system that listens for the enemy instead of using cameras is being developed, indicating a technology maturity signal. [SIG-974818D83206]
- A new drone detection system listens for the enemy instead of using cameras, capable of detecting drones hiding behind buildings or flying through fog. [SIG-974818D83206]
- Drone threats are forcing a new look at US airport security. [SIG-69003C1E5931]

## What Orange would deliver, and why it can win

### What Orange would deliver

Orange would assemble a solution starting with the Orange Drone Guardian offer, which provides drone detection and mitigation. This is enhanced by Orange's AI/Data/Cloud experts to deploy machine learning models for acoustic detection, and Cybersecurity experts to ensure the system is secure. Defense and security experts tailor the solution for

airport or defense environments. The engagement would begin with a site assessment, then deploy acoustic sensors (likely third-party) integrated with Orange's platform, and finally provide a monitoring service with alerts.

### Why Orange can win

Orange can win by leveraging its Orange Drone Guardian offer, which is a dedicated drone detection solution, combined with its sovereign infrastructure and cybersecurity expertise. The ability to provide a trusted, sovereign service is a differentiator, especially for defense and airport customers. However, the evidence does not specify Orange's existing acoustic detection capabilities, so the advantage may be more in integration and trust than in the core sensor technology.

### Named assets behind this — each individually inspectable (LK-08)

| Asset                        | Type            | Link | Why it is linked                                           | Curator     |
|------------------------------|-----------------|------|------------------------------------------------------------|-------------|
| Orange Drone Guardian        | offer           | L0   | offer addresses the use case AND provides the technology   | unconfirmed |
| AWS                          | partner         | L2   | partner provides the required technology                   | unconfirmed |
| Google Cloud                 | partner         | L2   | partner provides the required technology                   | unconfirmed |
| NICE                         | partner         | L2   | partner provides the required technology                   | unconfirmed |
| NVIDIA                       | partner         | L2   | partner provides the required technology                   | unconfirmed |
| AI / Data / Cloud experts    | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Cybersecurity experts        | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Digital experts              | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Defense and security experts | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Healthcare experts           | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Orange Group researchers     | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |

## Competitive landscape

**MEDIUM** competitive intensity (score 7.0; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

| Competitor       | Category                     | Position here                                                                                      | Basis      | Evidence                           |
|------------------|------------------------------|----------------------------------------------------------------------------------------------------|------------|------------------------------------|
| Verizon Business | Telecom operator peer        | competes in Cybersecurity                                                                          | evidenced  | SIG-AE409C225269, SIG-35ECED6AF5BB |
| AWS              | Hyperscaler                  | sells Machine learning; competes in Cloud — also an Orange partner                                 | structural | —                                  |
| Google Cloud     | Hyperscaler                  | sells Machine learning; competes in Cloud — also an Orange partner                                 | structural | —                                  |
| Atos / Eviden    | Global systems integrator    | sells Machine learning; active in Aerospace and defense industry; competes in Cloud, Cybersecurity | structural | —                                  |
| NICE             | Customer-experience platform | sells Machine learning — also an Orange partner                                                    | structural | —                                  |
| Databricks       | Data and AI platform         | sells Machine learning; competes in Cloud                                                          | structural | —                                  |
| Dataiku          | Data and AI platform         | sells Machine learning; competes in Cloud                                                          | structural | —                                  |
| Palantir         | Data and AI platform         | sells Machine learning; active in Aerospace and defense industry; competes in Cloud                | structural | —                                  |

**evidenced** means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

## Qualifying questions for the first meeting

- 1 Have you experienced a drone incursion that disrupted operations in the past year?
- 2 What is your current drone detection approach, and where are its gaps (e.g., fog, line-of-sight)?

- 3 Are you under regulatory pressure to deploy drone detection by a specific date?
- 4 How do you handle false positives from existing detection systems?
- 5 What is your preference for on-premises vs. cloud-based detection analytics?
- 6 Who in your organization owns airspace security decisions?

### Objections you will meet

| They will say                                                          | You can answer                                                                                                                                                                                                    |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| We already have camera-based drone detection; why do we need acoustic? | Cameras are effective in clear conditions, but they fail in fog or when drones are behind obstacles. Acoustic detection complements them, providing a second layer that covers those blind spots.                 |
| Lockheed Martin and Verizon already offer a drone detection service.   | That service relies on existing 5G towers, which may not cover your entire airspace. Orange can provide a dedicated, sovereign solution tailored to your site, with the assurance of a trusted European operator. |
| Is acoustic detection proven in noisy airport environments?            | It's an emerging technology, but the evidence shows it can detect drones in conditions where cameras fail. We would run a pilot at your site to validate performance before full deployment.                      |

### Next action, by role (FR-17)

| Role                   | Do this next                                                                                                                                                                                                                                                                                                                        |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presales / Proposal    | Bid brief: for a defense or airport tender, assemble a differentiated proposal around Orange Drone Guardian, highlighting its acoustic detection capability and the backing of AI / Data / Cloud and Defense and security experts, and consider partnerships with AWS, Google Cloud, or NVIDIA to strengthen the technical stack.   |
| Sales                  | In a conversation with a European airport operator's head of security, say: 'Our Orange Drone Guardian can detect drones even when they're hidden behind buildings or flying through fog, because it listens for them rather than relying on cameras—would you be interested in a pilot to see how it fits your airspace security?' |
| Strategist / Innovator | Deep-dive brief: assess whether to invest in maturing Orange Drone Guardian (L0) toward an L1 offer by studying acoustic drone-detection use cases in airport and defense settings, leveraging AI / Data / Cloud experts and Defense and security experts.                                                                          |

## Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

| Signal           | Date       | Publisher                    | T | Title and link                                                                                     |
|------------------|------------|------------------------------|---|----------------------------------------------------------------------------------------------------|
| SIG-974818D83206 | 2026-08-16 | bing.com                     | 2 | Forget cameras, this new drone detection system just listens for the enemy instead                 |
| SIG-69003C1E5931 | 2026-08-13 | thetraveler.org              | 2 | Drone Threats Force New Look at US Airport Security - thetraveler.org                              |
| SIG-AE409C225269 | 2026-08-13 | lightreading.com             | 2 | Lockheed Martin preps drone detection service using Verizon 5G                                     |
| SIG-35ECED6AF5BB | 2026-08-12 | bing.com                     | 2 | Lockheed Martin converts Verizon 5G into subscription drone alerts for critical infrastructure     |
| SIG-37D6B973F20E | 2026-08-12 | bing.com                     | 2 | Lockheed Martin demonstrates AI-powered drone detection system using 5G networks                   |
| SIG-CBC989FFDC45 | 2026-08-11 | JFeed                        | 2 | Unauthorized Drone Enters Ben Gurion Airport Airspace - JFeed                                      |
| SIG-28F5E44DC63D | 2026-07-27 | Fortune Business Insights    | 2 | Drone Detection Market Size, Share   Growth Report [2026-2034] - Fortune Business Insights         |
| SIG-4812FDBC16E3 | 2026-07-13 | airportindustry-news.com     | 2 | How Airports Can Plan Drone Detection for Low-Altitude Airspace Operations - airportindustry-ne... |
| SIG-2F0861536F6A | 2026-06-23 | Bao VietNamNet               | 2 | Drone detection system trialed at Da Nang International Airport - Bao VietNamNet                   |
| SIG-ABA568B306B8 | 2026-06-16 | Qazinform                    | 2 | Vietnam airports to install drone detection systems by 2027 - Qazinform                            |
| SIG-2CC796AF2DBE | 2026-06-01 | Unmanned airspace            | 2 | Lithuania deploys drone detection system at Palanga Airport, Kaunas and Vilnius to follow - Unm... |
| SIG-50BA4C222C4C | 2026-06-01 | International Airport Rev... | 2 | Palanga Airport deploys advanced drone detection system to strengthen airspace security measure... |
| SIG-17F9020F9769 | 2026-05-29 | The Baltic Times             | 2 | Palanga Airport installs advanced drone detection system: residents are urged to respect rules ... |
| SIG-86EDC2F80FE3 | 2026-05-07 | PRWeb                        | 2 | VIRTOWER Enhances Airspace Safety With Remote ID Drone Detection Upgrade - PRWeb                   |

## How this brief was made

**Computed, not written.** Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

**Written, then checked.** The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: what\_is\_changing (named entities that were never supplied: Lockheed Martin); competitive\_landscape (named entities that were never supplied: Lockheed Martin); risks\_and\_unknowns (named entities that were never supplied: Lockheed Martin); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

| What                        | Version                     | When                      |
|-----------------------------|-----------------------------|---------------------------|
| Scores — weight set         | w-2026-08-a                 | 2026-08-18T19:08:26+00:00 |
| Market sizing — assumptions | size-2026-08-a              | 2026-08-18T21:06:14+00:00 |
| Competitor register         | competitors-2026-08-a       | 2026-08-18T21:06:17+00:00 |
| Narrative — prompt / model  | describe-v1 / deepseek-chat | 2026-08-19T04:56:54+00:00 |
| Pipeline                    | 0.1.0                       | 2026-08-18                |

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.